BIOCOMPATIBILITY OF MATERIALS

Asst. Prof. Sumama Nuthana kalva

INTRODUCTION

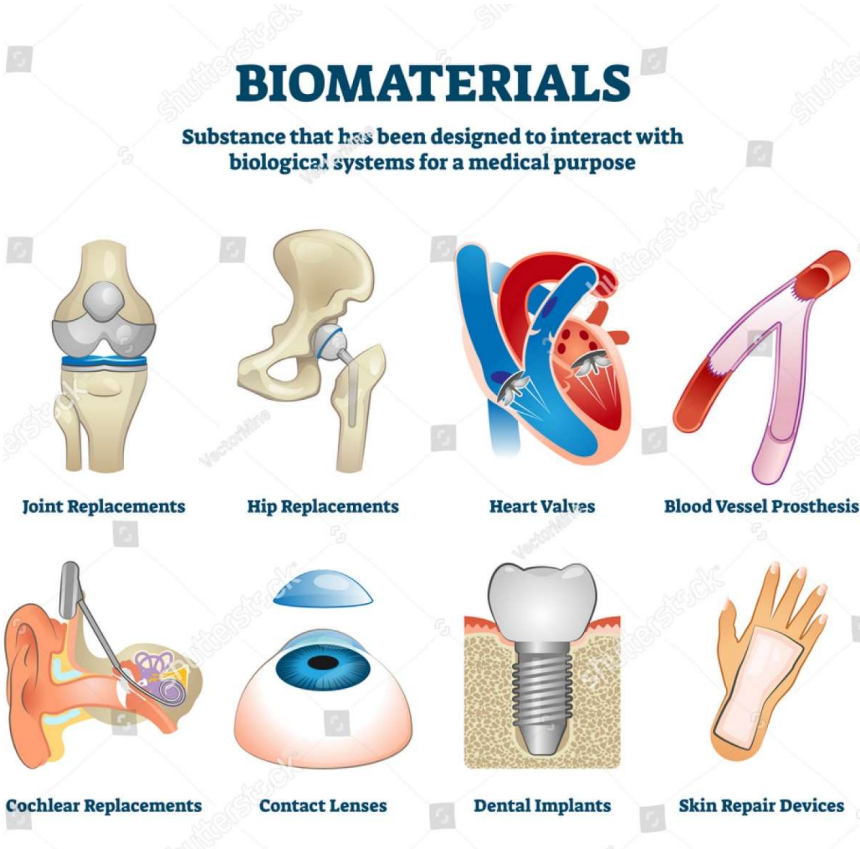
- **Biological Interactions with Materials** refer to the study of how biological entities (cells, tissues, organs) interact with synthetic or natural materials used in medical applications.

Importance in Biomedical Engineering:

- Critical for the successful integration of implants and devices.
- Ensures functionality, longevity, and patient safety.
- Guides the development of novel biomaterials with optimized interactions.

Applications:

- Implants: Orthopedic, cardiovascular, dental.
- Drug Delivery Systems: Nanoparticles, liposomes.
- Tissue Engineering: Scaffolds for regenerative medicine.



What is biocompatibility?

- Biocompatibility - ability of material to elicit an appropriate biological response on a given application in the body.

- The ability of a material to perform with an appropriate host response in a specific application", Williams' definition.

- "The quality of not having toxic or injurious effects on biological systems".

NEED FOR BIOCOMPATIBILITY TESTING

Medical devices and their component materials may leach compounds or have surface characteristics that may produce undesirable effects when used clinically. The selection and evaluation of materials and devices intended for use in humans requires a structured program of assessment to establish biocompatibility and safety.

Current regulations, whether in accordance with the U.S. Food and Drug Administration (FDA), the International Organization for Standardization (ISO), or the Japanese Ministry of Health, Labor and Welfare (JMHLW), require that manufacturers conduct adequate safety testing of their finished devices through pre-clinical and clinical phases as part of the regulatory clearance process.

Standards in Biomaterials Testing

Technical Committee 194 of the International Organization for Standardization (ISO) meet every spring

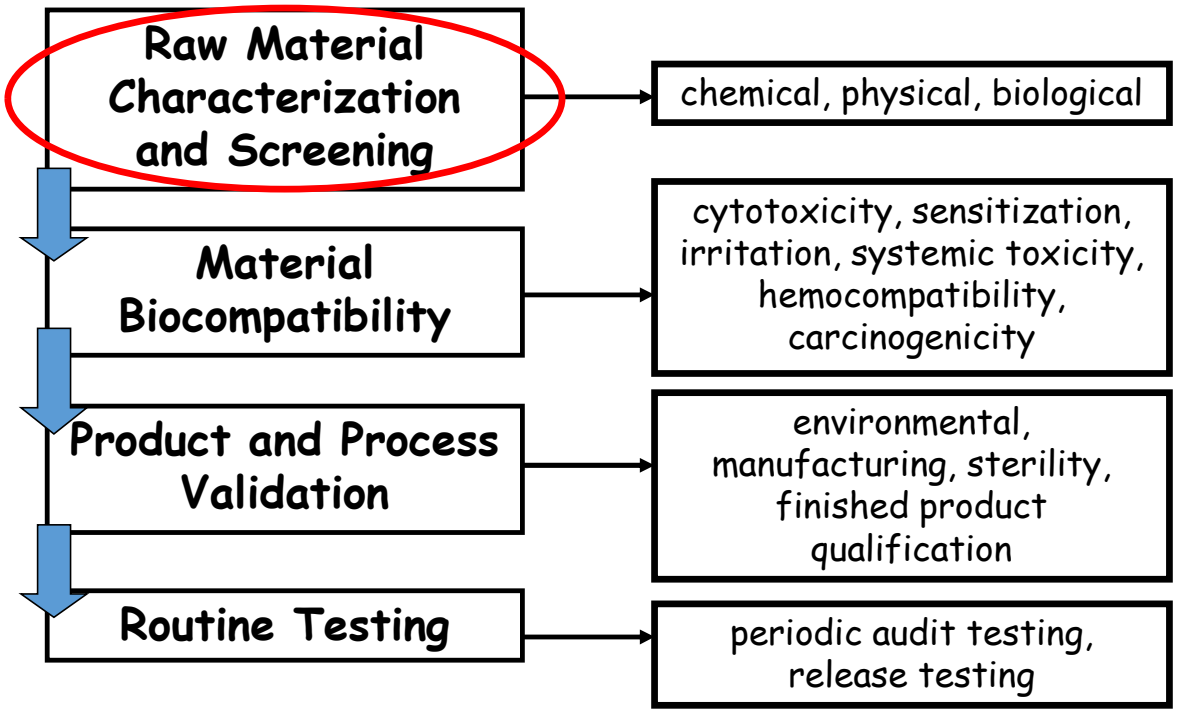
- Set of documents 10993 (FDA's version #G95-1):
  - 10993-1: "Guidance on Selection of Tests."
  - 10993-2: "Animal Welfare Requirements."
  - 10993-3: "Tests for Genotoxicity, Carcinogenicity, and Reproductive Toxicity."
  - 10993-4: "Selection of Tests for Interactions with Blood."
  - 10993-5: "Tests for Cytotoxicity—In Vitro Methods."
  - 10993-6: "Tests for Local Effects after Implantation."
  - 10993-7: "Ethylene Oxide Sterilization Residuals."
  - 10993-9: "Degradation of Materials Related to Biological Testing."
  - 10993-10: "Tests for Irritation and Sensitization."
  - 10993-11: "Tests for Systemic Toxicity."
  - 10993-14: "Materials Evaluation."

ISO 10993-1: Introduction to Standards

Principles of toxicity evaluation

1. Characterize and identify the composition of biomaterial, potential impurities and extractables
2. Account for toxic effects of leachable chemicals and degradation products
3. Testing performed by competent and informed persons
4. Data available to reviewing authorities
5. Reevaluate toxic effects if there are changes in composition, manufacturing practice or intended use
6. Account for all data: nonclinical, clinical, postmarket etc.
7. All materials should undergo cytotoxicity, sensitization and irritation tests

Testing Biomaterials: Four Phases



ISO 10993-14: Materials Characterization

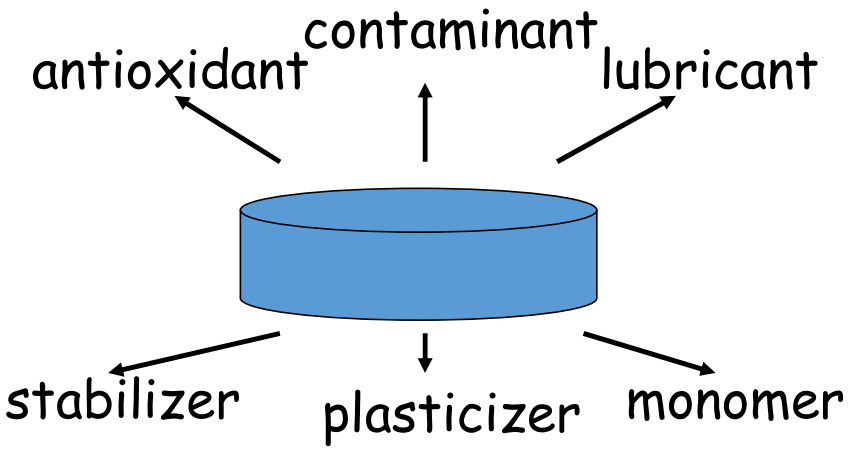
Chemical, toxicological, physical, electrical, morphological and mechanical properties:

- material
- additives
- process contaminants and residues
- leachable substances
- degradation products

ISO 10993-14: Materials Characterization

Why characterize?

- To establish baseline fingerprint
- To determine the presence and nature of any extractable chemicals



ISO 10993-14: Materials Characterization

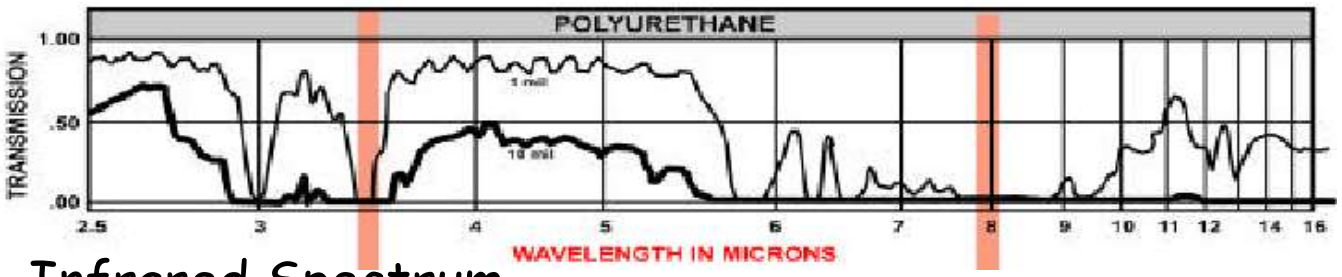
Chemical fingerprint

- Infrared spectroscopy
- Raman spectroscopy
- Thermal analysis (melting point, degree of crystallinity, glass transition)

Spectroscopy: every chemical bond displays unique vibrational energy

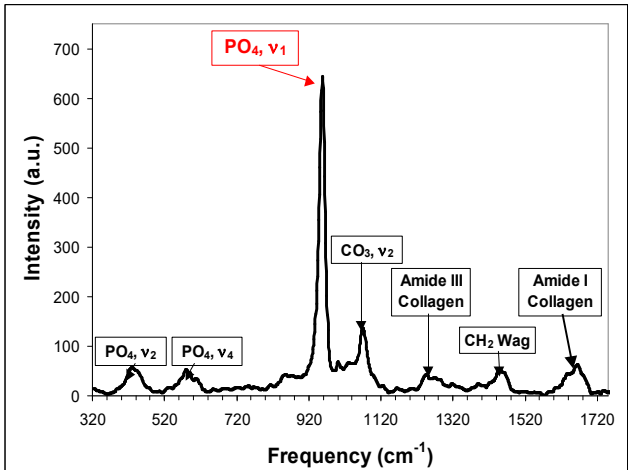
- excite with laser
- monitor the vibrational spectrum

ISO 10993-14: Materials Characterization



Infrared Spectrum of Polyurethane

Raman Spectrum of Bone

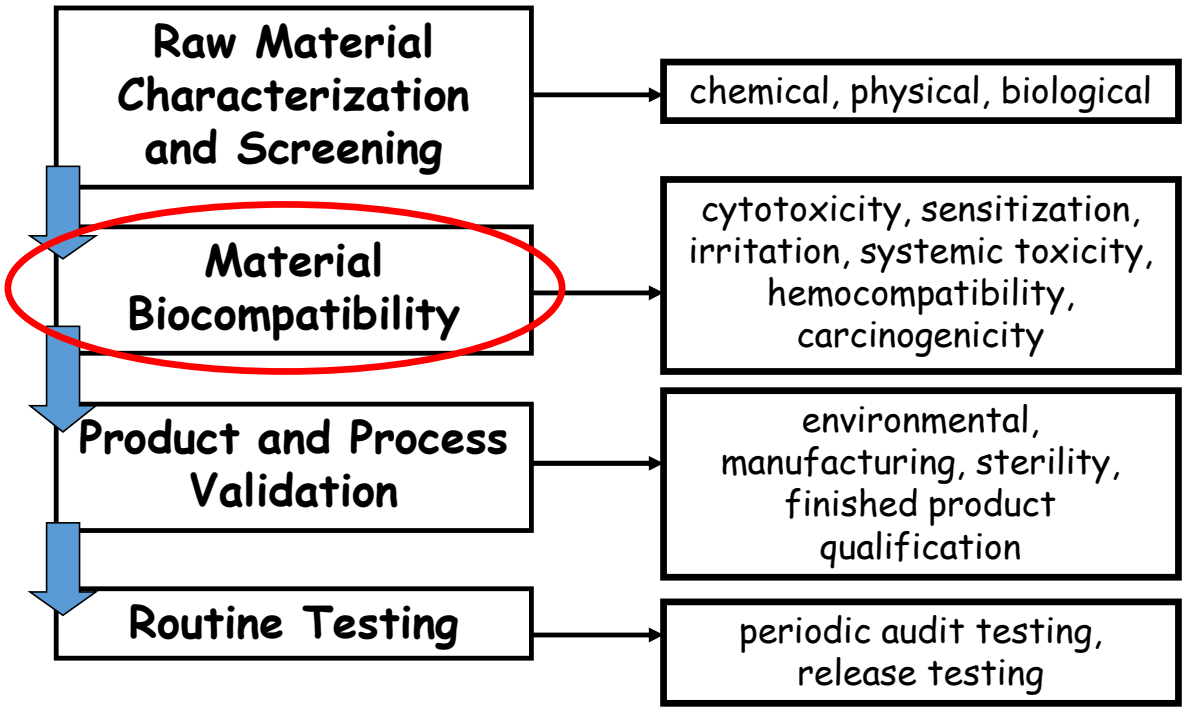


ISO 10993-14: Materials Characterization

Mechanical tests

- Tension, compression, bending
- Fracture toughness
- Fatigue

Testing Biomaterials: Four Phases



Guidelines

Three levels for biocompatibility testing

- **Level 1** Series of in vitro studies and preclinical test in animals. These studies can be used as a screening phase but should be followed with more detailed investigation (Exposure)

- **Level 2** In usage test in animals at the sites where the materials have to be used.

- **Level 3** Clinical trial in humans, after the data have been sufficiently screened in the first two levels

Testing Biomaterials

- ***IN VITRO*** (cell cultures in glass)
  - rapid
  - inexpensive
  - poor representation of physiological conditions
  - good as the first step
- ***IN VIVO*** (animal experiments)
  - better approximation to human environment
  - demanding protocols (Animal Welfare Act)
  - right animal model approximate human environment
  - second step prior to clinical use

ISO 10993-5: CYTOTOXICITY

Definition: The degree to which a substance can cause harm to living organisms.

Mechanisms of Toxicity:

- **Direct Cytotoxicity:** Direct damage to cells upon contact.
- **Indirect Effects:** Triggering of inflammatory pathways or oxidative stress.



Sources of Toxicity in Biomaterials:

- Breakdown of materials over time can release potentially harmful substances.
- Residual chemicals from manufacturing processes.
- Molecules that detach from the material surface.

Mitigation Strategies:

- **Material Selection:** Choosing inherently non-toxic materials.
- **Surface Coatings:** Applying biocompatible coatings to prevent leaching.
- **Purification Processes:** Ensuring high purity during manufacturing.

In vitro Testing

- Cell cultures
  - dissect tissue
  - finely chop
  - free cells from extracellular matrix via digestion (trypsin, collagenase etc.)
  - place them in a dish with nutrients and keep them warm!

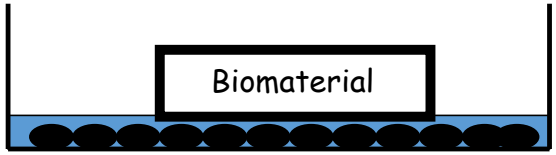


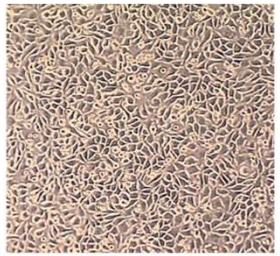
In vitro Testing

- Assay methods:
  - ❖ Direct contact
  - ❖ Agar diffusion
  - ❖ Elution

In vitro Testing: Direct Contact

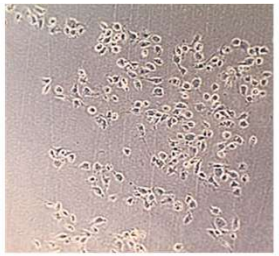
- Direct contact
  - monolayer, confluent cell culture, L-929 mouse fibroblasts
  - biomaterial in direct contact
  - 24 hours, 37 ° C
  - cells may
    - change morphology
    - die
    - lose adherence to dish
  - hematoxylin blue: stains live adherent cells
  - toxicity=dead/live !
- Why L-929?
  - easy to maintain
  - good correlation with animals tests
  - fibroblasts present in wound healing





A confluent monolayer (100 x magnification) of well-defined L929 mouse fibroblast cells exhibiting cell-to-cell contact.

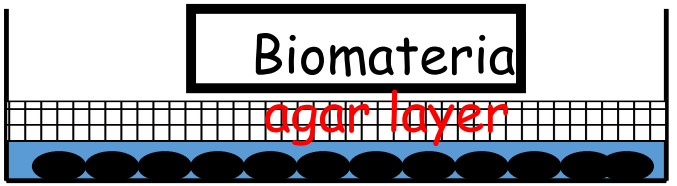
This appearance is indicative of a non-cytotoxic (negative) response

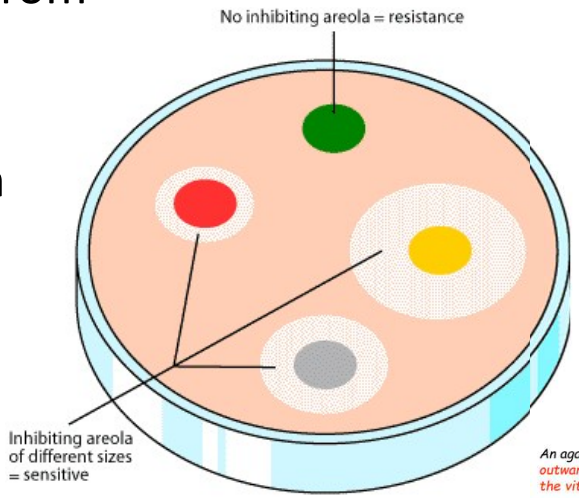


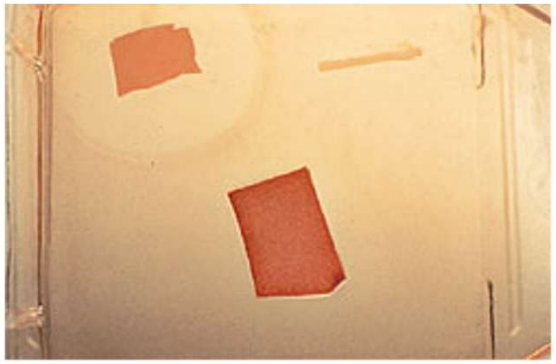
L929 mouse fibroblast cells (100 x magnification) that illustrate a positive cytotoxic reaction in the elution test method. The cells are grainy and lack normal cytoplasmic space; the considerable open areas between cells indicate that extensive cell lysis (disintegration) has occurred.

In vitro Testing: Agar Diffusion

- Agar diffusion
  - agar layer between cells and biomaterial
  - agar: gel-like polymer derived from red algae
  - chemicals diffuse through agar
  - use special stain to label health cells
  - areola of unstained dead cells around the biomaterial



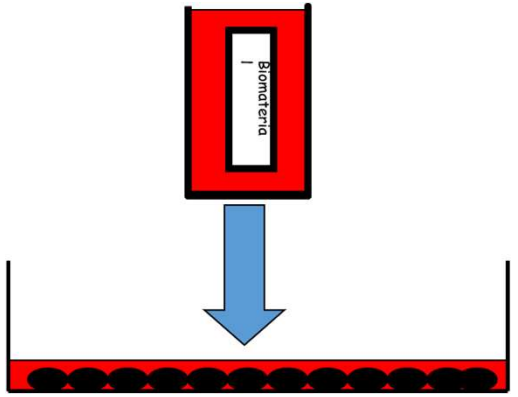


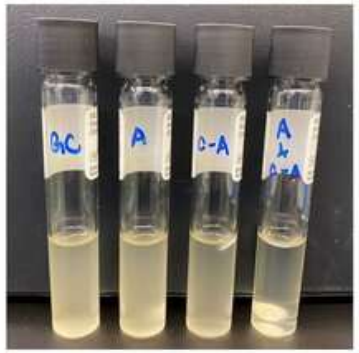


An agar diffusion flask containing a sample of positive control material. The discoloration that extends outward from the material indicates that the presence of the sample has caused the cells to lyse, losing the vital stain incorporated in the agar layer.

In vitro Testing: Elution

- Elution
  - prepare extract of a material
  - how? keep the material in oil based or water based solution (*Why oil or water?*)
  - chemicals will leach into solution
  - apply solution to cell-culture
  - perform similar stain based viability tests





In vitro Testing: Comparison

TABLE 1 Advantages and Disadvantages of Cell Culture Methods

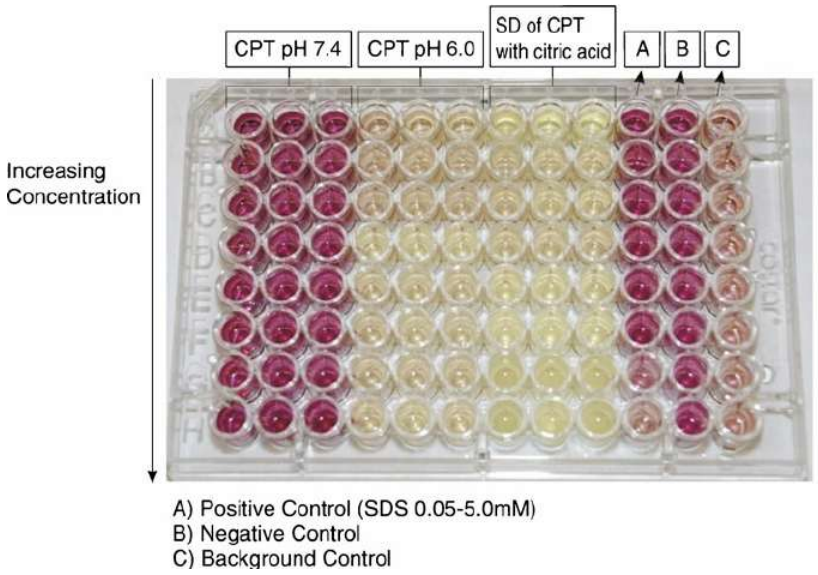
	Direct contact	Agar diffusion	Elution
Advantages	Eliminate extraction preparation Zone of diffusion Target cell contact with material Mimic physiological conditions Standardize amount of test material or test indeterminate shapes Can extend exposure time by adding fresh media	Eliminate extraction preparation Zone of diffusion Better concentration gradient of toxicant Independent of material density Use filter paper disk to test liquids or extracts	Separate extraction from testing Dose response effect Extend exposure time Choice of extract conditions Choice of solvents
Disadvantages	Cellular trauma if material moves Cellular trauma with high density materials	Requires flat surface Solubility of toxicant in agar Risk of thermal shock when preparing agar overlay Limited exposure time Risk of absorbing water from agar	Additional time and steps

CYTOTOXICITY

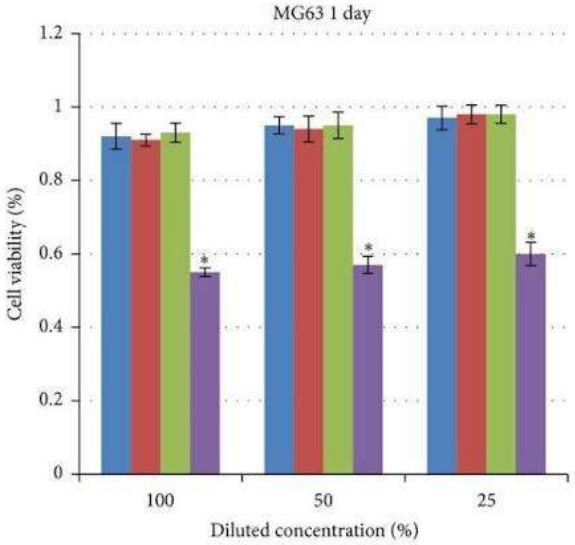
Testing Methods:

1. MTT Assay:

- **Principle:** Measures mitochondrial activity as an indicator of cell viability.
- **Procedure:** Reduction of MTT reagent to formazan by metabolically active cells



A) Positive Control (SDS 0.05-5.0mM)
B) Negative Control
C) Background Control



CYTOTOXICITY

2. Live/Dead Staining:Principle:

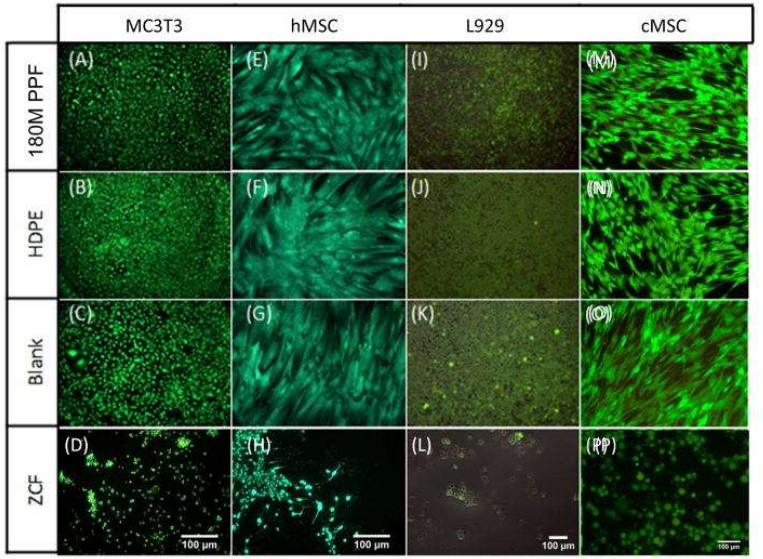
- Differentiates live cells (stained with calcein AM) from dead cells (stained with ethidium homodimer).
- **Application:** Visual assessment of cell viability under a microscope.

3. LDH Assay

- **Principle:** Measures lactate dehydrogenase released from damaged cells.
- **Application**: Quantitative assessment of cell membrane integrity.

4. Annexin V/PI Staining:

- **Principle:** Differentiates apoptotic and necrotic cells.
- **Application:** Detailed analysis of cell death pathways.



Interpretation of Results:

- **High Cytotoxicity**: Indicates potential for material rejection or tissue damage.
- **Low/No Cytotoxicity**: Suggests suitability for implantation.

In vivo Testing

Critical for development of clinical devices

***In vitro* tests cannot replace *in vivo* tests**:

- no inflammation
- no immune response
- single cell type
- no tissue remodeling

In vivo tests provide:

- interactions of different cell types
- effects of hormonal factors
- interactions with extracellular matrix
- interactions with blood-borne cells, proteins and molecules

ISO 10993-10: HYPERSENSITIVITY

Definition: ***An exaggerated or inappropriate immune response to a material, leading to adverse effects*.**



Hypersensitivity reactions

	Type I	Type II	Type III	Type IV
Mediators	IgE	IgG/ IgM	Antigen-antibody complex	T cells
Antigen form	Soluble antigen	Cell-bound antigen	Soluble antigen	Soluble/cell bound antigen
Mechanism of action	Allergen-specific IgE causes mast-cell degranulation releasing histamine and other inflammatory mediators	IgG/IgM antibodies bind to cell antigens, leading to complement activation and destruction of the cell	Antigen-antibody complexes are deposited in tissues where they activate complement chain causing local inflammation and recruitment of leucocytes	Antigen-presenting cells activate Th1/cytotoxic T cells. T-cells recruit macrophages and release inflammatory cytokines
Time course	Immediate-phase: minutes Late-phase: 8-12 hours	Hours to days	Hours to days	Days (delayed)
Examples	• Anaphylaxis • Hay fever • Food allergies • Drug allergies • Eczema • Asthma	• Acute transfusion reaction • Haemolytic disease of the newborn	• Rheumatoid arthritis • Post-streptococcal glomerulonephritis • Systemic lupus erythematosus	• Contact dermatitis • Tuberculin test • Graft rejection

HYPERSENSITIVITY

Risk Factors:

- **Material Composition**: Metals (e.g., nickel, cobalt) are common allergens.
- **Surface Properties**: Surface roughness and porosity can influence immune recognition.
- **Patient Sensitivity**: Pre-existing allergies or immune conditions.

Prevention and Management:

- **Material Selection:** Using hypoallergenic materials (e.g., titanium, platinum).
- **Surface Passivation:** Coatings that shield immunogenic components.
- **Patient Screening**: Identifying individuals with known sensitivities.
- **Immunomodulatory Strategies**: Incorporating anti-inflammatory agents or immunosuppressants.

ISO 10993-10: Sensitization

Albino Guinea pigs: as responsive to dermal sensitizers as human beings

Test methods:

- repeated patch (Buehler): for topical devices such as dermal electrodes and surgical gowns
  - Induction phase: expose shaved back directly to material under occlusive dressings. 6 hours/day, 3 days/week, 3 weeks
  - Recovery phase: 2 weeks rest to allow for development of response
  - Final exposure
- maximization (Magnuson-Kligman): used for materials that will contact areas other than the skin
  - fluid extracts of test material prepared in saline or vegetable oil
  - inject extract with an adjuvant agent that will enhance immune response
  - two weeks rest
  - apply extract topically

ISO 10993-10: Sensitization



Positive response to maximization test in a guinea pig.



Sensitization (rash) to latex glove after use for several weeks or months.

http://www.devicelink.com/mddi/archive/98/05/025.html

ISO 10993-10: Irritation

Irritation: local tissue response characterized by the usual signs of inflammation

- redness
- swelling
- heat
- pain

Chemicals

- additives
- processing
- manufacturing aids (e.g. detergent residue, ethylene oxide residue)
- inadvertent contaminants

ISO 10993-10: Irritation

In vivo tests for irritation:

- Intracutaneous (like allergy tests)
- primary skin
- ocular

Intracutaneous test:

- albino rabbits
- prepare fluid extract under controlled temperature, duration, material surface/volume ratio (water and oil based solvent)
- extract injected into the skin
- multiple sites
- inject controls
- observe for evidence of redness and swelling at 24h, 48h, 72h
- aggressive test, extract prepared under exaggerated conditions
- maximizes the chance of finding irritant chemical



Positive response to maximization test in a albino rabbit.

ISO 10993-10: Irritation

Primary skin test

- less aggressive than intracutaneous
- placement of material on shaved back of albino rabbits
- cover with occlusive dressing
- apply between 4-24 hrs
- observe for 72 hrs
- score for redness and swelling
- compare with known values for primary skin irritation
- categorize the response: negligible, slight, moderate, severe



ISO 10993-10: Irritation

Ocular test:

- used for eye contact products
- fluid extracts (occasionally solids or powders)
- placed directly into the pocket of the lower eyelid of an albino rabbit
- other eye untreated, control
- observe regularly up to 72 hours
- score based on:
  - swelling and redness of conjunctiva
  - response of iris to light
  - corneal opacity
  - presence of discharge

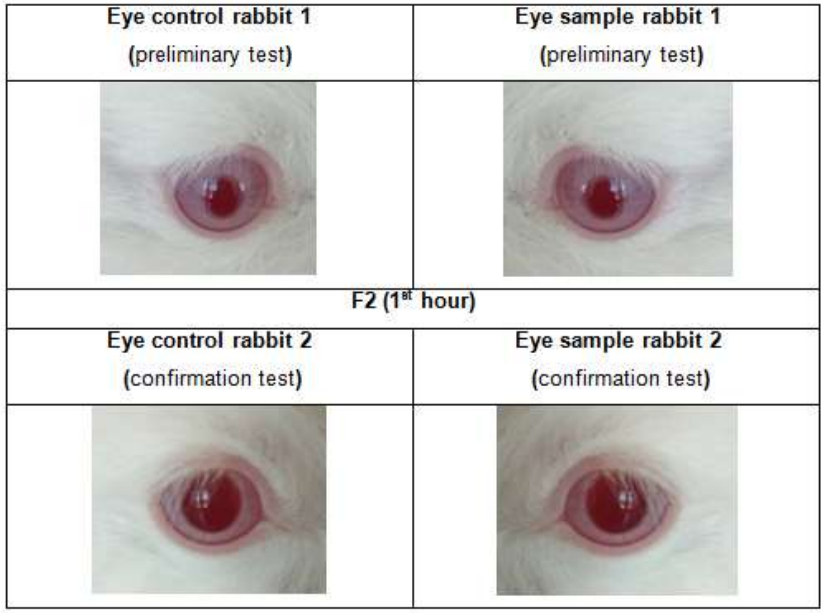


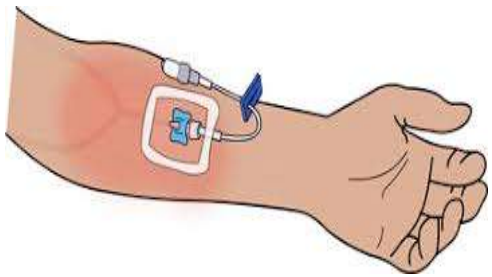
Fig 3: Example image of the result of irritation test on rabbit's eye

INFLAMMATION

- A complex biological response of body tissues to harmful stimuli, such as pathogens, damaged cells, or foreign materials (biomaterials).
- A protective mechanism aimed at eliminating the initial cause of cell injury, removing damaged tissue, and initiating tissue repair.
- When biomaterials are implanted, the body may recognize them as foreign, triggering an inflammatory response.
- The nature and extent of inflammation influence the success or failure of the implant.
- Two types: acute and chronic inflammation







INFLAMMATION - TYPES

a. Acute Inflammation

- Rapid onset, typically within minutes to hours.
- Short duration, resolving within days.
- Examples: sore throat, burns, insect bite
- Clinical Signs: **Redness (rubor), heat (calor), swelling (tumor), pain (dolor), and loss of function.**
- Cellular Involvement: Predominantly neutrophils (polymorphonuclear leukocytes) and Activation of the complement system and release of pro-inflammatory cytokines (e.g., TNF-α, IL-1β).
- Role in Biomaterial Response: Initial response to implantation. Essential for clearing debris and preventing infection but excessive acute inflammation can lead to complications.







INFLAMMATION - TYPES

b. Chronic Inflammation

- **Characteristics:**
  - Persistent inflammation lasting weeks to months or longer.
  - Can result from unresolved acute inflammation or ongoing exposure to irritants.
- **Clinical Signs:**
  - Less pronounced redness and heat.
  - Presence of lymphocytes, macrophages, and plasma cells.
  - Tissue destruction and attempts at repair (fibrosis, granuloma formation).
- **Cellular Involvement:**
  - Macrophages, lymphocytes, fibroblasts.
  - Release of different cytokines (e.g., IL-6, TGF-β).
- **Role in Biomaterial Response:**
  - Indicates a sustained foreign body reaction. It Can lead to fibrosis and encapsulation of the implant, impairing functionality.



Psoriasis



Rheumatoid Arthritis



Allergy

EFFECTS OF INFLAMMATION ON PERFORMANCE

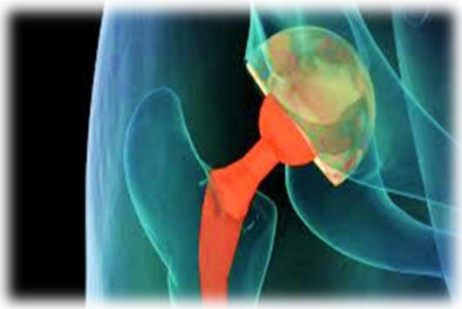
a. Integration vs. Rejection

- **Successful Integration:**
  - Controlled, minimal inflammation facilitates tissue integration and healing.
  - Promotes vascularization and incorporation of the material into the host tissue.
- **Rejection and Failure:**
  - Excessive or chronic inflammation leads to fibrous encapsulation or rejection.
  - Impairs the functionality and longevity of the implant.

b. Impact on Surrounding Tissue

- **Tissue Damage:**
  - Inflammatory mediators and ROS can damage surrounding cells and tissues.
  - Chronic inflammation may lead to fibrosis, osteolysis (in **orthopedic implants**), or other tissue-specific pathologies.
- **Functional Impairment:**
  - Encapsulation with fibrous tissue can isolate the implant, reducing its effectiveness (e.g., reduced insulin release from a glucose sensor).







MODULATING INFLAMMATORY RESPONSES

The inflammatory response to biomaterials is crucial in determining their biocompatibility and clinical success. Modulating this response means designing or modifying materials to either minimize harmful inflammation or promote beneficial immune responses.

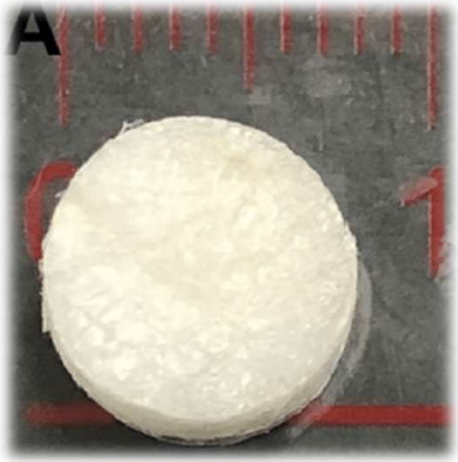
1. Material Design Strategies

a. Biomimicry

- Designing biomaterials to mimic the natural structure, composition, and properties of the extracellular matrix (ECM) to reduce immune recognition and promote tissue integration.

Example:

Collagen-based scaffolds: Collagen is a natural component of the ECM. Biomaterials designed with collagen, or its derivatives, can closely mimic the natural environment, reducing the immune response and encouraging cells to grow and integrate with the material. Collagen scaffolds are often used in wound healing and tissue engineering.



Collagen based scaffolds

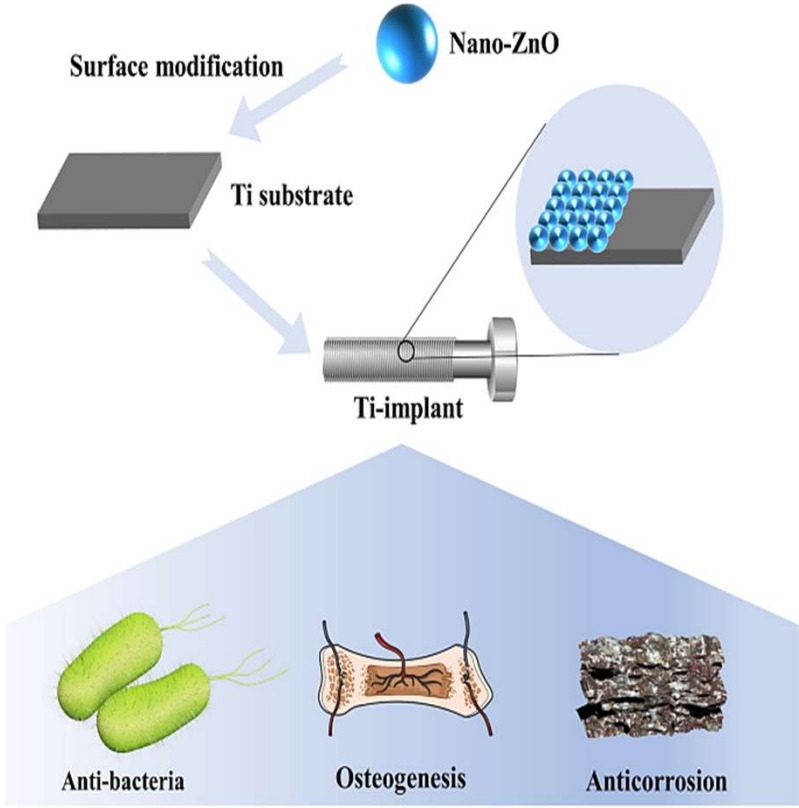
MODULATING INFLAMMATORY RESPONSES

b. Surface Engineering

- Altering the surface properties (e.g., chemistry, topography) of biomaterials to modulate protein adsorption and cell behavior, which in turn affects inflammation.

Example:

- **Nanostructured titanium implants:** Surface nano-patterning of titanium used in orthopedic and dental implants has been shown to reduce the inflammatory response by influencing how proteins and cells interact with the surface. Nanostructures promote better osseointegration and reduce inflammatory cytokine production from immune cells.



MODULATING INFLAMMATORY RESPONSES

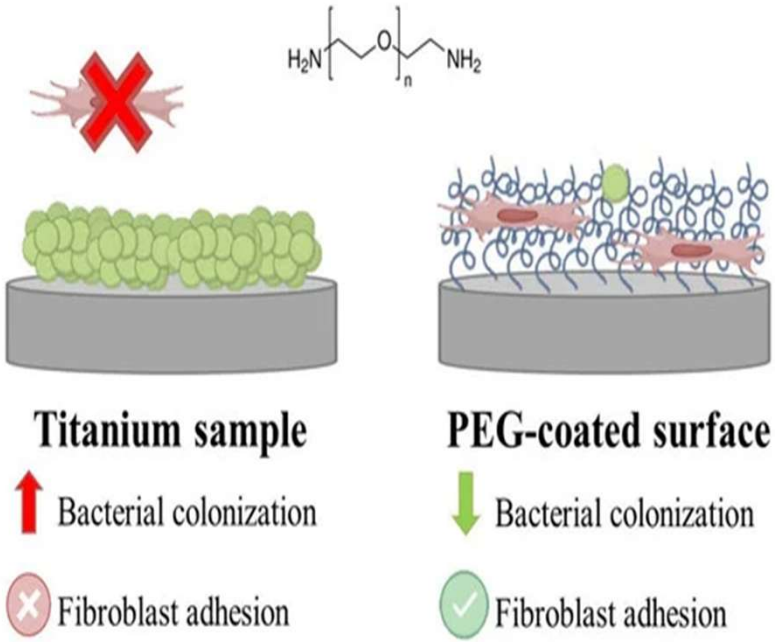
2. Surface Modifications

a. Coatings

- Applying bioactive coatings on the surface of biomaterials to reduce immune activation, limit protein adsorption, and prevent inflammatory reactions.

Example:

- **Polyethylene Glycol (PEG) Coatings:** PEG is a hydrophilic polymer that, when coated on biomaterials, resists protein adsorption (creating a "stealth" surface), which reduces the recognition by immune cells. This makes the material less prone to initiating an immune response.



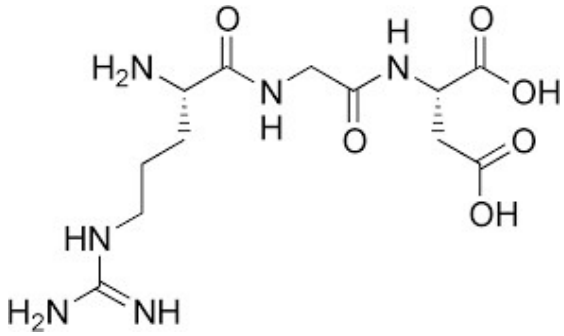
MODULATING INFLAMMATORY RESPONSES

b. Functionalization

- Attaching bioactive molecules or functional groups to the surface of biomaterials that promote an anti-inflammatory environment by interacting with specific immune pathways.

Example:

- **Arginylglycylaspartic acid (RGD) peptides:** These peptides are known for promoting cell adhesion. When functionalized on biomaterial surfaces, RGD peptides can enhance tissue integration and reduce inflammatory signaling by encouraging tissue-friendly interactions between cells and the material.
- **Outcome:** Implants functionalized with RGD peptides have been shown to promote faster wound healing and better tissue regeneration, while reducing prolonged inflammation.



MODULATING INFLAMMATORY RESPONSES

3. Use of Anti-inflammatory Agents

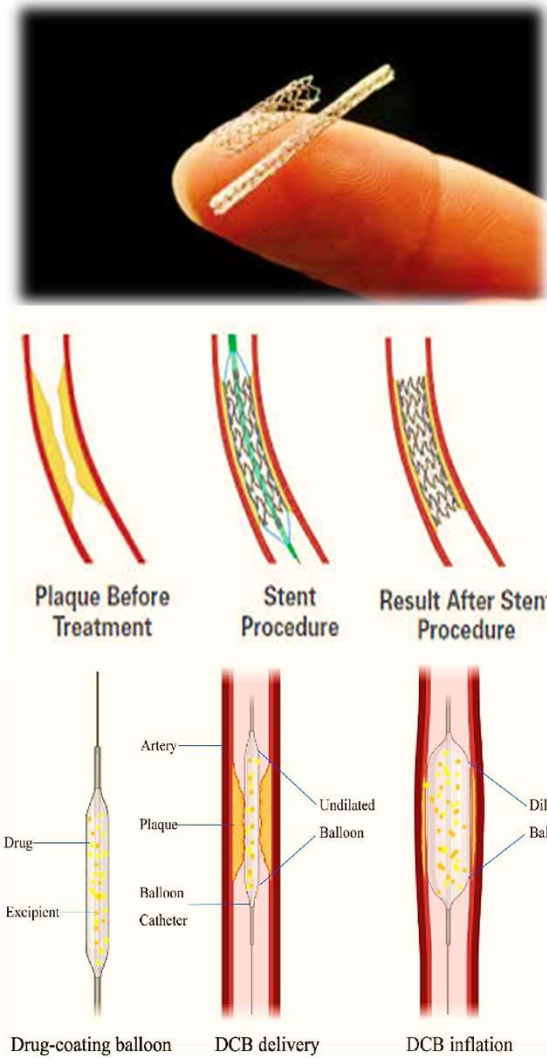
a. Drug-Eluting Implants

- Biomaterials can be designed to release anti-inflammatory drugs over time to control and reduce inflammation at the site of implantation.

Example:

- **Corticosteroid-eluting stents:** Coronary stents used in heart patients can be coated with corticosteroids or other anti-inflammatory drugs to prevent excessive inflammation and restenosis (re-narrowing of the artery) after implantation.
- **Outcome:** Drug-eluting stents have significantly reduced post-implantation complications such as chronic inflammation and fibrosis, improving the long-term outcomes of heart patients.

Example: Paclitaxel-coated implants used in cancer treatments reduce local inflammation and prevent excessive tissue proliferation, which can lead to implant failure.



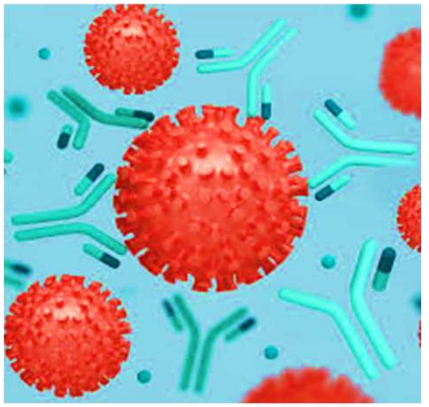
MODULATING INFLAMMATORY RESPONSES

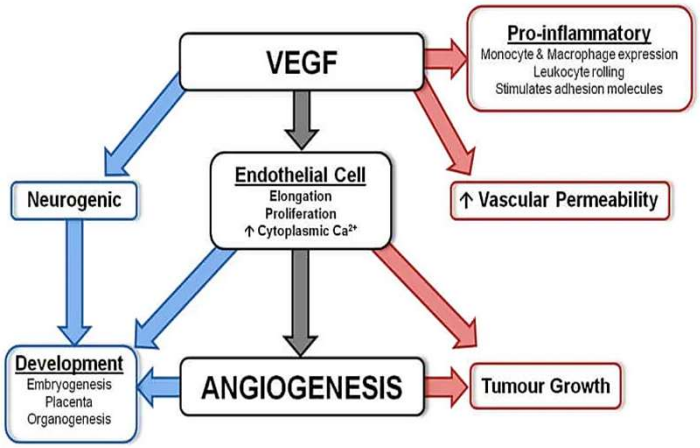
b. Biologics

- The use of biologic agents, such as cytokines, antibodies, or growth factors, to modulate the immune system's response to the biomaterial.

Example:

- **Anti-TNF-α antibodies:** These antibodies can be delivered locally at the site of biomaterial implantation to block the action of tumor necrosis factor-alpha (TNF-α), a pro-inflammatory cytokine that plays a major role in chronic inflammation.
- **Outcome:** Local delivery of anti-TNF-α antibodies has shown to significantly reduce inflammation around implants, especially in chronic cases like arthritis where biomaterial implants are used to replace damaged joints.
- **Example: VEGF-functionalized scaffolds** (Vascular Endothelial Growth Factor) have been used to promote tissue vascularization while simultaneously reducing inflammatory responses.





MODULATING INFLAMMATORY RESPONSES

4. Immunomodulatory Biomaterials

- Materials specifically designed to actively modulate the immune response, shifting it towards tissue repair and regeneration instead of prolonged inflammation.

Example:

- **Hydrogels with anti-inflammatory properties:** Some hydrogels can be designed to release anti-inflammatory cytokines like IL-4, promoting the polarization of macrophages into the M2 phenotype (which is involved in tissue repair rather than inflammation).
- **Outcome:** Hydrogels that release IL-4 have been shown to reduce chronic inflammation and improve wound healing outcomes in tissue engineering applications.







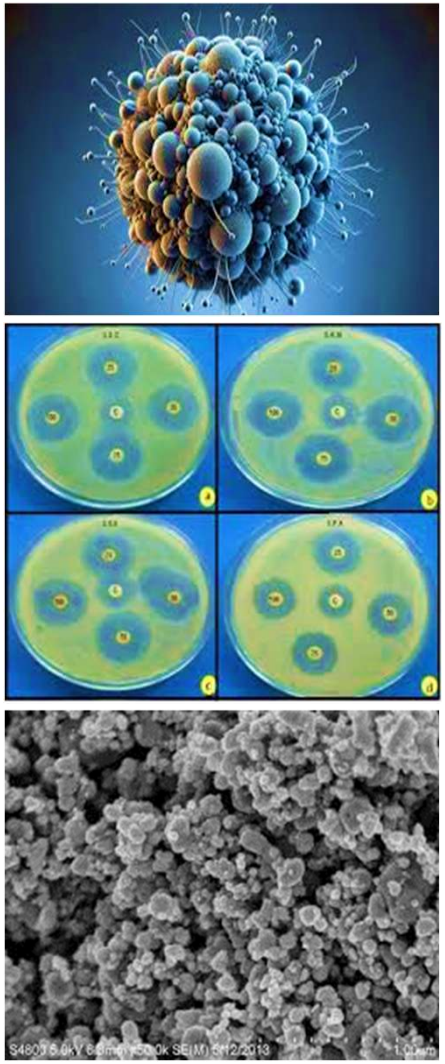
MODULATING INFLAMMATORY RESPONSES

5. Nanotechnology Approaches

- Nanoparticles (NPs) can be incorporated into biomaterials or used as carriers for delivering drugs or biologics that modulate the immune system.

Example:

- **Silver nanoparticles:** Silver nanoparticles (AgNPs) are widely known for their antimicrobial properties. When incorporated into biomaterials, they can also modulate inflammation by reducing the bacterial load at the implant site, which would otherwise induce a strong inflammatory response.
- **Outcome:** Silver nanoparticle-infused wound dressings have been shown to reduce inflammation, prevent infection, and promote faster healing, making them effective for burn patients.



ISO 10993-11: Systemic Effects

- Effects of released chemicals on liver, heart, kidneys, and brain
- Mice and rats used generally
- Various routes of application
  - dermal
  - inhalation
  - intravenous
  - intraperitoneal
  - oral

ISO 10993-11: Systemic Effects

Application:

- fluid extracts (intraperitoneal or intravenous)
- implantation of material (particularly biodegradable ones) (intramuscular, intraperitoneal, subcutaneous)

- Collect
  - blood samples (hematology, serum chemistry)
  - tissue samples (pathology)
- Observe adverse signs
  - convulsions
  - prostration
  - weight loss

ISO 10993-11: Systemic Effects

Persistence

- acute (within 24 hours)
- sub-acute (14-28 days)
- sub-chronic (up to 90 days or more than 10% of animal's life span)
- chronic (more than 90 days or more than 10% of animal's life span)

ISO 10993-6: Implant Effects

- Most direct means of evaluating a medical device material's effect on the surrounding tissue
- Mice, rats, guinea pigs, rabbits
- Implant cut to size, sterilized, and implanted aseptically
- Attention focused on local effects

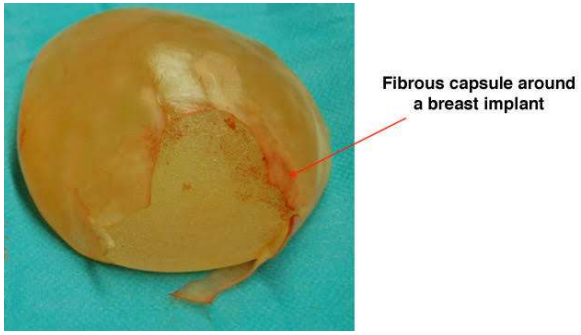
ISO 10993-6: Implant Effects

Rabbits:

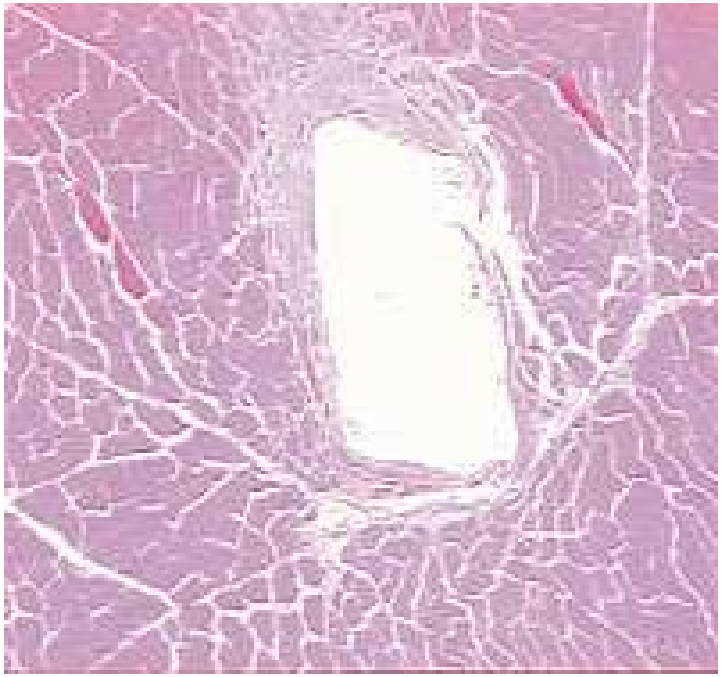
- 1x10 mm strips of material
- sterilized
- placed in gauge needles
- anesthetics
- four test samples and four plastic (inert to body) controls implanted in the paralumbar muscle
- Test samples and controls on opposite sides
- short term: tissue response observed after 1-12 weeks
- long term: tissue response observed after 12-78 weeks

ISO 10993-6: Implant Effects

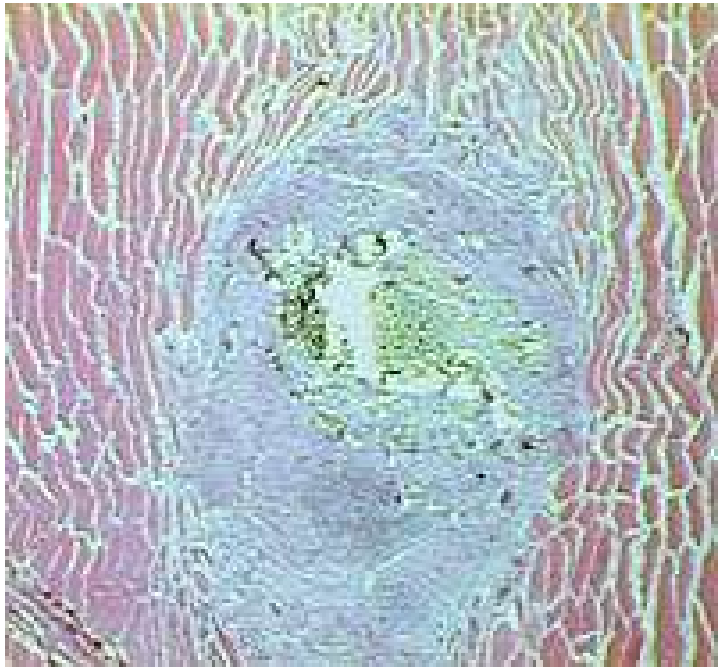
- At each interval observe the size of fibrous capsule
  - reactive material: 2-4 mm thick
  - no visible capsule for control material
- Histopathological examination
  - inflammatory reaction to the implant
  - cell death around implant



ISO 10993-6: Implant Effects



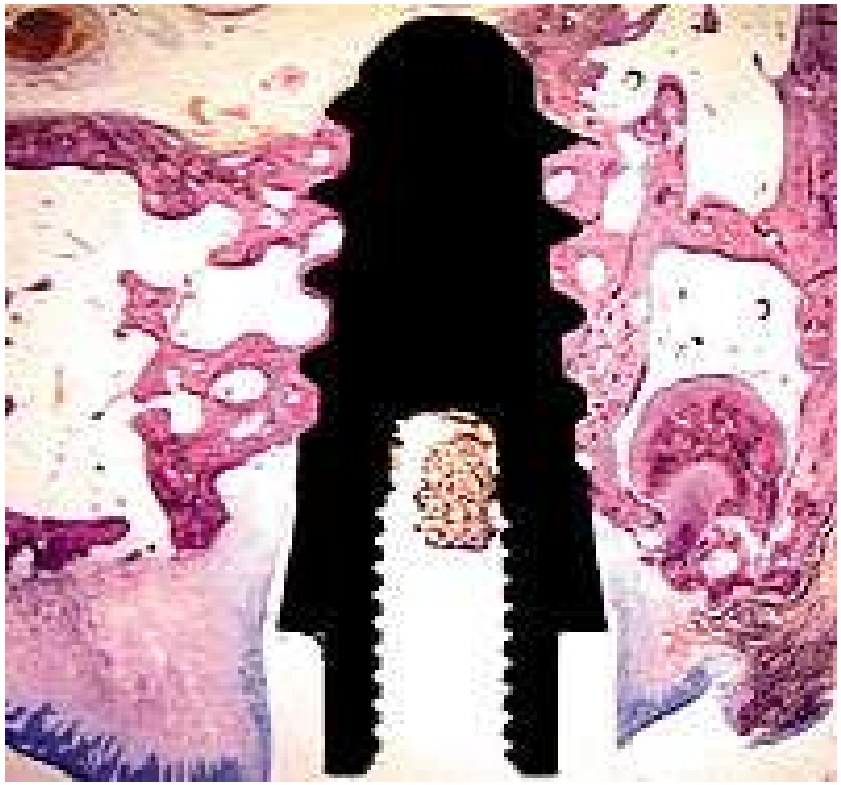
MINIMAL
RESPONSE



SEVERE
RESPONSE

http://www.devicelink.com/mddi/archive/98/08/013.html

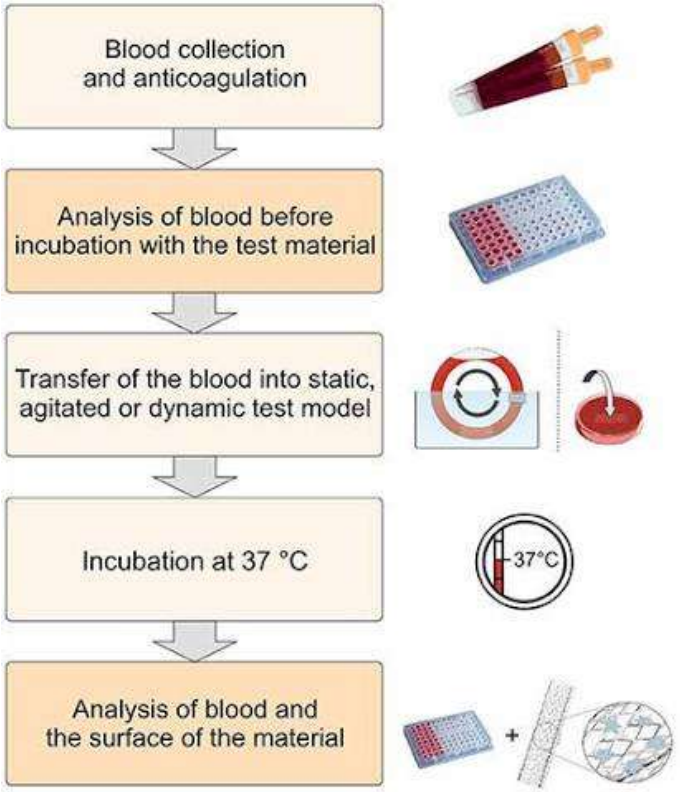
ISO 10993-6: Implant Effects



http://www.devicelink.com/mddi/archive/98/08/013.html

ISO 10993-4: Hemocompatibility

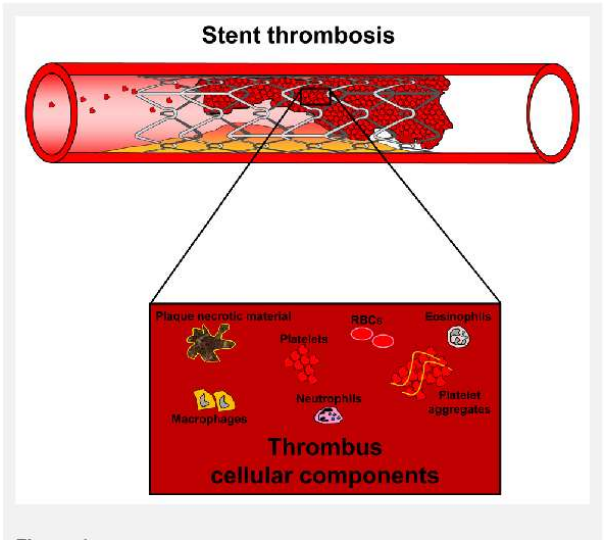
- Blood has multitude of important cells types and proteins:
  - ❖oxygen carrying erythrocytes (viability)
  - ❖antigen specific lymphocytes (immune)
  - ❖white blood cells (inflammation)
  - ❖coagulation proteins
- Mechanical or material mediated damage to cells
- In vitro and in vivo tests possible
- In vitro tests require the use of anticoagulants



ISO 10993-4: Hemocompatibility

SAMPLE OF TESTS CONDUCTED FOR TESTING HEMOCOMPATIBILITY

Test category	Method	Comments
Thrombosis	Adhered platelets, leukocytes, aggregates, erythrocytes, fibrin, etc.	Light microscopy or scanning electron microscopy
Coagulation	Partial thromboplastin time	
Platelets	Platelet count	
Hematology	Leukocyte count; hemolysis (plasma hemoglobin)	Hemolysis is performed to measure the red blood cell membrane fragility in contact with materials and devices.
Immunology	C3a, C5a, TCC, Bb, iC3b, C4d, SC5b-9	



CARCINOGENICITY

Definition: The potential of a material or its degradation products to induce cancer.

- Long-term implants must not contribute to oncogenesis.
- Certain monomers, additives, or degradation products may be carcinogenic.
- Chronic exposure increases risk.
- Higher concentrations of carcinogenic substances elevate risk
- Metabolic processes may activate or deactivate carcinogens.

Assessment of Carcinogenicity

- **Cell Transformation Assays:** Assessing changes in cell morphology and growth.
- **Genotoxicity Tests:** Evaluating DNA damage (e.g., Ames test, comet assay).
- **Animal Models:** Long-term implantation studies to monitor tumor formation.
- **Epidemiological Studies:** Tracking cancer incidence in patients with implants.

CARCINOGENICITY

Regulatory Requirements:

- **IARC Classification:** International Agency for Research on Cancer guidelines.
- **FDA and ISO Standards**: Specific testing protocols for carcinogenic potential.

Examples of Carcinogenic Concerns:

- **Polyurethane:** Degradation can release diisocyanates, potential carcinogens.
- **Certain Phthalates:** Used as plasticizers, linked to endocrine disruption and cancer.
- **Metals:** Some metal ions (e.g., chromium VI) are known carcinogens.

Mitigation Strategies:

- **Material Modification:** Using non-carcinogenic alternatives.
- **Stabilization:** Preventing degradation through cross-linking or other chemical modifications.
- **Encapsulation:** Isolating potentially harmful components within biocompatible coatings.

Regulatory and Ethical Considerations:

- **Pre-Market Testing:** Comprehensive evaluation before clinical use.
- **Post-Market Surveillance:** Ongoing monitoring for adverse effects in patients.

Genotoxicity

Genetic toxicology tests are used to investigate materials for possible mutagenic effects, that is, damage to the body's genes or chromosomes.

In any living organism, the smallest unit capable of independent existence is the cell, and each cell has specific functions that it must maintain to sustain life. Instructions for these activities are encoded within genes, or sections of DNA, devoted to each specific function. A thread consisting of many genes strung together with large amounts of DNA is called a chromosome.

An alteration in any part of this DNA structure that results in permanent inheritable changes in cell function is called a mutation, and the agents that cause such mutations are known as genotoxic agents or genotoxins

A mutation is a change in the formation content of the genetic material (DNA code) that is propagated through subsequent generations of cells. Mutations can be classified into two general types: gene mutations and chromosomal mutations. Gene mutations are changes in nucleotide sequences at one or several coding segments within a gene; chromosomal mutations are morphological alterations or aberrations in the gross structure of the chromosomes.

BIOCOMPATIBILITY TESTING

[Based on ISO 10993-1:2009 and FDA G95-1 Guidelines]

DEVICE CATEGORIES		Contact Duration A – Limited (≤ 24 hrs) B – Prolonged (>24 hrs to ≤30 days) C – Permanent (>30 days)	BIOLOGICAL EFFECT: Initial: Cytotoxicity	Sensitization	Irritation	Systemic Toxicity (Acute)	Subchronic Toxicity (Subacute)	Genotoxicity	Implantation	Hemocompatibility	Other*: Chronic Toxicity	Carcinogenicity
Body Contact												
SURFACE DEVICES	Skin	A	●	●	●							
		B	●	●	●							
		C	●	●	●							
	Mucosal Membranes	A	●	●	●							
		B	●	●	●	◊	◊		◊			
		C	●	●	●	◊	●	●	◊		◊	
	Breached or Compromised Surfaces	A	●	●	●	◊						
		B	●	●	●	◊	◊		◊			
		C	●	●	●	◊	●	●	◊		◊	
EXTERNAL COMMUNICATING DEVICES	Blood Path, Indirect[1]	A	●	●	●	●				●		
		B	●	●	●	●	◊			●		
		C	●	●	◊	●	●	●	◊	●	◊	◊
	Tissue[1]/Bone/ Dentin Communicating	A	●	●	●	◊						
		B	●	●	●	●	●	●	●			
		C	●	●	●	●	●	●	●		◊	◊
	Circulating Blood[1]	A	●	●	●	●		◊[2]		●		
		B	●	●	●	●	●	●	●	●		
		C	●	●	●	●	●	●	●	●	◊	◊
IMPLANT DEVICES	Tissue / Bone	A	●	●	●	◊						
		B	●	●	●	●	●	●	●			
		C	●	●	●	●	●	●	●		◊	◊
	Blood[1]	A	●	●	●	●	●		●	●		
		B	●	●	●	●	●	●	●	●		
		C	●	●	●	●	●	●	●	●	◊	◊

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[1] "Tissue" includes tissue fluids and subcutaneous spaces.

[2] For all devices used in extracorporeal circuits.

[3] Pyrogenicity / Materials Mediated should be considered.

[4] Based on the risk assessment, additional tests may be necessary.

● – ISO Evaluation Tests to Consideration

◊ – Additional tests that the FDA considers may be applicable.

For reproductive and biodegradation tests, contact your WuXi AppTec Account Manager.

Test	< 0.5mm thickness (120cm²/20mL ratio) *[film, sheet, tubing wall]*	≥ 0.5mm thickness (60cm²/20mL ratio) *[tubing wall, slab, molded items]*	Irregularly Shaped (4g/20mL ratio) *[powder, pellets, foam, non-absorbent molded items]*	Membranes (2g/20mL ratio) *(low-density materials)*	Liquids
CYTOTOXICITY					
MEM Elution Using L-929 Cells – ISO/USP *[18 days]*	1 x 30cm²	1 x 15cm²	1 x 1g	1 x 0.5g	3mL
Extract Colony by Elution – JMHLW *[31 days]*	120cm² or 2g				10mL
Agarose Overlay Using L-929 Cells – ISO/USP *[14 days]*	Sufficient material to produce 3 patches, 1cm x 1cm each				1mL
MTT Cytotoxicity Assay with L-929 Mouse Fibroblast Cells *[26 days]*	1 x 36cm²	1 x 18cm²	1 x 1.2g	1 x 0.8g	NA
HEMOCOMPATIBILITY / BLOOD COMPATIBILITY					
Hemolysis: ASTM – Direct Contact *[21 days]*	≤ 0.5mm thick: 3 x 42cm² • > 0.5mm thick: 3 x 21cm² • Irregularly shaped: 3 x 1.4g				21mL
Hemolysis: ASTM – Extract *[21 days]*	≤ 0.5mm thick: 3 x 60cm² • > 0.5mm thick: 3 x 30cm² • Irregularly shaped: 3 x 2g				NA
Hemolysis: NIH – Direct Contact *[21 days]* • NIH – Extract *[21 days]*	5g/10mL ratio: 3 x 2g • 0.5g/10mL ratio (low density materials): 3 x 0.2g				NA
Hemolytic Toxicity – JMHLW *[32 days]*	3 x 90cm²	3 x 45cm²	3 x 3g	3 x 1.5g	NA
Complement Activation C3a and SC5b-9 *[25 days]*	1 x 6cm²	1 x 3cm²	1 x 0.2g	NA	NA
Partial Thromboplastin Time *[25 days]*	3 x 4cm² (test samples) and 3 x 4cm² (comparison samples)				
Platelet and Leukocyte Count *[25 days]*	3 x 12cm² (test samples) and 3 x 12cm² (comparison samples)				
In Vitro Hemocompatibility *[25 days]*	3 x 12cm² (test samples) and 3 x 12cm² (comparison samples)				
Thrombosis (*In Vivo*) – 2 Dog *[28 days]* *Other animal models available.*	2 test samples and 2 comparison samples [Length: 8-15cm, Maximum Diameter: ≤ 3mm]				
GENOTOXICOLOGY					
Bacterial Mutagenicity (Ames) *[28 days]*	2 x 24cm²	2 x 12cm²	2 x 0.8g	2 x 0.4g	4mL
Bacterial Reverse Mutation – JMHLW *[40 days]*	5g *[Sample amounts will increase if residue is obtained during pretest.]*				
In Vitro Chromosome Aberration *[62 days]*	2 x 42cm²	2 x 21cm²	2 x 1.4g	2 x 0.7g	8mL
In Vitro Chromosome Aberration – JMHLW *[62 days]*	5g *[Sample amounts will increase if residue is obtained during pretest.]*				
In Vitro Mouse Lymphoma *[39 days]*	2 x 42cm²	2 x 21cm²	2 x 1.4g	2 x 0.7g	7mL
In Vivo Mouse Micronucleus *[55 days]*	2 x 120cm²	2 x 60cm²	2 x 4g	2 x 2g	18mL

IRRITATION / STIMULATION					
Intracutaneous Reactivity – ISO/USP/JMHLW *[29 days]*	2 x 36cm²	2 x 18cm²	2 x 1.2g	2 x 0.6g	5mL
Vaginal Mucosal Irritation – ISO *[54 days]*	10 x 60cm²	10 x 30cm²	10 x 2g	10 x 1g	30mL
Primary Skin Irritation – ISO *[30 days]*	Sufficient material to produce 7 patches, 2.5cm x 2.5cm each				5mL
Primary Skin Stimulatory – JMHLW *[30 days]*	2 x 48cm²	2 x 24cm²	2 x 1.6g	2 x 0.8g	NA
PYROGENICITY					
Material Mediated Rabbit Pyrogen – ISO and JMHLW *[26 days]*	900cm²	450cm²	30g	15g	150mL
SENSITIZATION					
Murine Local Lymph Node Assay (LLNA) *[32 days]*	6 x 30cm²	6 x 15cm²	6 x 1g	6 x 0.5g	3mL
Maximization Sensitization (Guinea Pig) – ISO *[54 days]*	6 x 60cm²	6 x 30cm²	6 x 2g	6 x 1g	15mL
Maximization Sensitization – JMHLW *[81 days]*	35g *[Sample amounts will increase if residue is obtained during pretest.]*				
Repeated Patch Dermal Sensitization Test – Buehler *[57 days]*	Sufficient material to produce 105 patches, 2.5cm x 2.5cm each				50mL
SYSTEMIC TOXICITY					
Acute Systemic Test – ISO/USP/JMHLW *[27 days]*	2 x 48cm²	2 x 24cm²	2 x 1.6g	2 x 0.8g	7mL
Subacute/Subchronic Toxicity (Mice) – 5 Dose Exposure *[89 days]*	5 x 30cm²	5 x 15cm²	5 x 1g	5 x 0.5g	*Inquire*
Subacute/Subchronic Toxicity (Mice) – 14 Dose Exposure *[89 days]*	14 x 30cm²	14 x 15cm²	14 x 1g	14 x 0.5g	*Inquire*
IMPLANTATION *[Note: TAT is in addition to implant duration.]*					
Intramuscular/Subcutaneous Implantation – ISO/JMHLW *[42 days]*	Sufficient material to produce 15 implants, approx. 10mm x 3mm				
Intramuscular Implantation – USP *[21 days]*	Sufficient material to produce 12 implants, approx. 10mm x 3mm				
CHEMISTRY					
Physicochemical Test for Plastics – USP *[7 days]*	10g or 600cm²				
FTIR *[7 days]*	5g				

BIOCOMPATIBILITY OF MATERIALS

Asst. Prof. Sumama Nuthana kalva